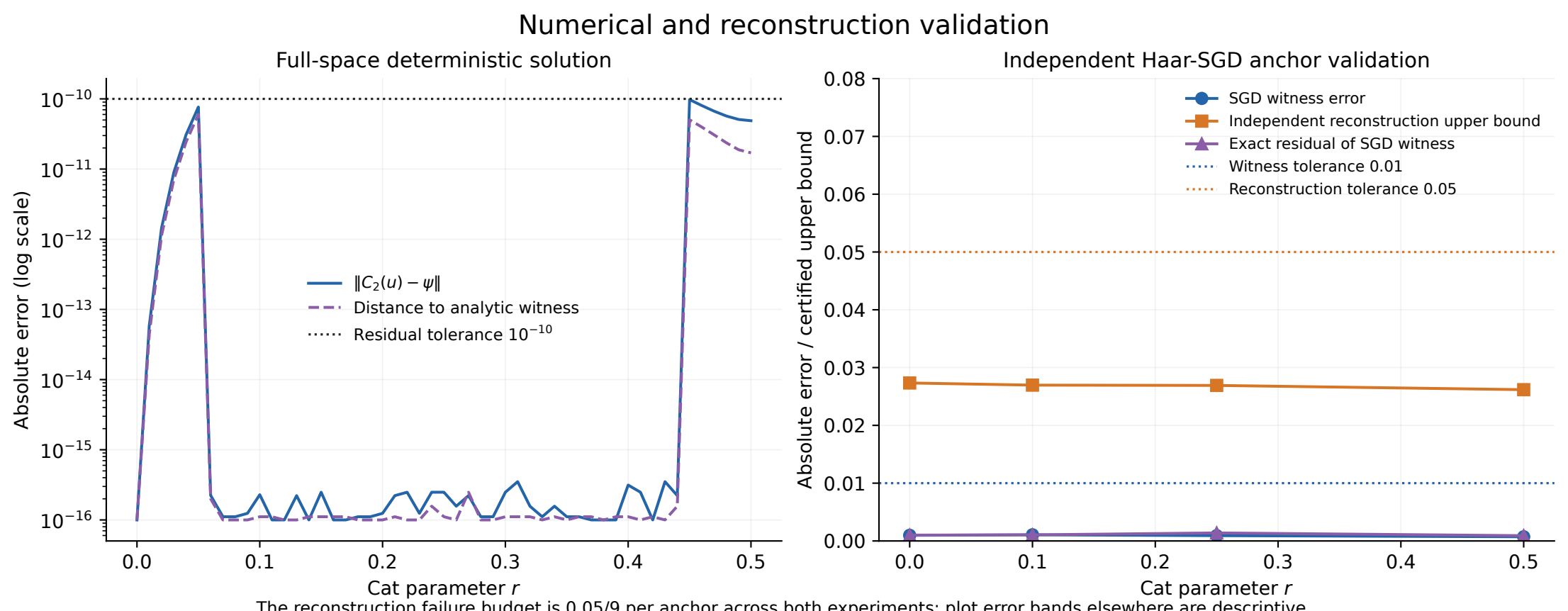
Numerical and reconstruction validation
Full-space deterministic solution
Absolute error (log scale)
10^{-10}
10^{-11}
10^{-12}
10^{-13}
10^{-14}
10^{-15}
10^{-16}
$\|C_2(u) - \psi\|$
Distance to analytic witness
Residual tolerance 10^{-10}
0.0
0.1
0.2
0.3
0.4
0.5
Cat parameter r
Independent Haar-SGD anchor validation
Absolute error / certified upper bound
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
SGD witness error
Independent reconstruction upper bound
Exact residual of SGD witness
Witness tolerance 0.01
Reconstruction tolerance 0.05
0.0
0.1
0.2
0.3
0.4
0.5
Cat parameter r
The reconstruction failure budget is 0.05/9 per anchor across both experiments; plot error bands elsewhere are descriptive.
Errors below 1e-16 are displayed at 1e-16 on the logarithmic axis.